Erdős Problems 249 and 257 in Lean 4:

formalised results, certificate reductions, and open questions

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Abstract

We formalise in Lean 4 several results around Erdős Problems 249 and 257. For every integer base $b \geq 2$, the development verifies the irrationality of the Erdős–Borwein series $\sum_{n\geq 1}(b^n - 1)^{-1}$, together with several structured infinite-support variants. For the totient constant $S = \sum_{n\geq 1} \varphi(n)/2^n$, whose irrationality remains open, we prove that any rational representation has denominator greater than $Q_0 \approx 7.96 \times 10^{34}$, and formalise an exact reduction of irrationality to pointwise non-integrality of binary tail differences, each equivalent to a finite decidable certificate. We also refine that reduction through an exact full-target interval, a squared-Mersenne enclosure, and a fresh-loss projection, and verify 28 explicit lcm-diagonal scales through $t = 64$. On the #257 side we formalise the measure-one, nowhere-dense Mersenne achievement set and auxiliary rationality criteria based on binary carry systems. For the test value $1/2$, membership is equivalent to infinitely many greedy skips; the upper last-producer branch and the first exceptional middle coordinate are impossible, leaving an exact tail estimate for the two remaining middle coordinates. This work solves neither Erdős #249 nor the universal form of Erdős #257.

Keywords. irrationality; Erdős–Borwein constant; Euler totient; Lambert series; binary carry systems; Lean 4; Mathlib; formal verification.

MSC 2020. 11J72 (primary); 11A25, 68V20 (secondary).

1 Introduction

Two open questions provide the setting; their original formulations occur in the Erdős–Graham problem collection and Erdős's later survey [4, 5], while the numbering follows the maintained Erdős Problems catalogue [22]. Erdős #249 asks whether the totient constant

$$S = \sum_{n\geq 1} \frac{\varphi(n)}{2^n}$$

is irrational. Erdős #257 asks whether $\sum_{n\in A}(2^n - 1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}$. The second problem contains the classical full-support Erdős–Borwein constant, but the passage from structured supports to an arbitrary infinite support remains open.

Across the two directions, exact tail and carry identities separate finite arithmetic, checked by Lean, from an analytic or combinatorial uniformity statement. This is a common proof architecture, not an assertion that the two open problems are equivalent. The present status is as follows.

problem	checked here	exact open edge
#249	Any rational expression $S = p/q$ has $q > Q_0 \approx 7.96 \times 10^{34}$. At fixed parameters, non-integrality of the relevant tail difference is equivalent to a finite certificate; 28 lcm-diagonal scales are verified through $t = 64$.	Produce these certificates at unbounded parameters. No such supply theorem is proved, so the irrationality of S remains open.
#257	The full-support series is irrational in every integer base $b \geq 2$, and several named infinite-support families are formalised. For the value $1/2$, membership is equivalent to infinitely many greedy skips; the upper last-producer branch and the middle coordinate -3 are eliminated.	Treat every infinite support. For the $1/2$ test value, the remaining checked endpoint is a tail estimate for the middle coordinates -2 and -1.

The checked library also contains identities, rigidity statements, finite certificates, and scoped no-go theorems. Their hypotheses and logical dependencies are stated explicitly. Neither the size of the library nor an unexplored composition is evidence that a solution is latent in it.

Reading map. Section 2 gives the common Mersenne–Lambert transform. Section 3 treats the #257 direction and begins its half-value analysis with a one-page account of the exact residual. Section 4 treats #249 and likewise begins with the certificate argument in one page. Sections 5 and 6 describe the formalisation and its verification; the appendices contain auxiliary carry criteria and the source cross-reference. Small blue source marks may be ignored on a first reading. The separate transport-and-curvature companion develops the technical affine and first-harmonic material without changing the open status of #249.

The prior-art boundary is theorem-specific. Chowla had conjectured the broader rational-argument irrationality question for $\sum_{n\geq 1} x^n/(1-x^n)$; the full-support specialization is $x = 1/b$. Erdős proved that integer-reciprocal case [2, 1]; Borwein later proved a broader irrationality theorem that includes the relevant Mersenne–Lambert specialisation [6]. The pairwise-coprime support condition is also classical [3]. Periodic-coefficient Lambert series have a separate literature [14]. Kovač–Tao record the strict-tail geometry used by the achievement-set appendix [15], while Wang–Grau Ribas use the integral carry recurrence in the weighted binary setting [16]. For adjacent work on the binary expansion of the Erdős–Borwein constant, see Crandall and the subsequent result of Campbell [17, 18]; neither result is used or formalised in this development. We make no priority claim for these mathematical ingredients; the contribution here is their verified integration, the stated certificate reductions and bounds, and the formalisation lessons described below.

The Mersenne–Lambert ladder. $L(f) = \sum_{n\geq 1} \frac{f(n)}{2^n - 1}$

$L(\mu) = \frac{1}{2}$	$L(\varphi) = 2$	$L(\mathbf{1}) = E$	$L(\varphi * \mu) = S$ (#249)
rational	rational	**irrational (proved)**	**open**

Logical status of the #249 reduction.

$S \in \mathbb{Q} \implies$ eventual binary periodicity $\implies R_{N+h} - R_N \in \mathbb{Z}$ for the eventual period.

$\mathsf{certifiedKill}(h, N, L) \implies R_{N+h} - R_N \notin \mathbb{Z}$, and
$\exists L\, \mathsf{certifiedKill}(h, N, L) \iff R_{N+h} - R_N \notin \mathbb{Z}$.

Solid implications and the equivalence are proved; the unresolved step is an unbounded supply of such non-integral tail differences.

2 The Mersenne–Lambert ladder

Define the Lambert-type transform of an arithmetic function f by

$$L(f) \;=\; \sum_{n\ge 1}\frac{f(n)}{2^n-1} \;=\; \sum_{m\ge 1}\frac{(f*\mathbf{1})(m)}{2^m}, \qquad (f*\mathbf{1})(m)=\sum_{d\mid m} f(d),$$

the second equality being the standard Lambert rearrangement (one Dirichlet convolution by the constant $\mathbf{1}=\zeta$); Apostol supplies the classical arithmetic-function and Dirichlet-convolution background [7]; Merca introduced a modern Lambert-series factorisation theorem [8], and Merca–Schmidt give a synthesis/unification treatment [9]. Walking the ladder by convolution gives five values with rational, irrational, open, or transcendental status.

input f	value	mathematical status	treatment here
μ	$L(\mu)=1/2$	rational	formalised
φ	$L(\varphi)=2$	rational	formalised
$\mathbf{1}$	$L(\mathbf{1})=E$	irrational	formalised
$A=\varphi*\mu$	$L(A)=S$	open (#249)	identities and bounds formalised
Id	$L(\mathrm{Id})=\sum\sigma(m)/2^m$	transcendental	cited [12]

The weight $A=\varphi*\mu$ is the primitive-conductor count: $A(p)=p-2$ on primes, $A\ge 0$, and $A*\zeta=\varphi$. One convolution by ζ separates the open constant S from the rational 2, and one more separates rational from transcendental ($\varphi*\zeta=\mathrm{Id}$, $\mathrm{Id}*\zeta=\sigma$). So S is the rung one convolution from the machine-checked irrational E: the same series $\sum(\cdot)/(2^d-1)$, with the constant weight 1 replaced by the nonnegative weight A.

Proposition 2.1 (the rational rungs)**.** *$\sum_{d\ge1}\mu(d)/(2^d-1)=1/2$ and $\sum_{d\ge1}\varphi(d)/(2^d-1)=2$, exactly.*

Formalised: `MerLamLad` and `MerLamLad`; restated on the #249 series shape at `CerKer` and `CerKer`.

Proposition 2.2 (the positive lift, $L(A)=S$)**.** *With the primitive-conductor weight $A=\varphi*\mu$,*

$$\sum_{d\ge1}\frac{A(d)}{2^d-1} \;=\; \sum_{n\ge1}\frac{\varphi(n)}{2^n} \;=\; S.$$

Formalised: `CerKer`, with the weight A defined at `MerLamLad`. This places S in the same $\sum(\cdot)/(2^d-1)$ family as E, with a nonnegative weight.

2.1 Finite primitive-coordinate obstruction

The weight A also has finite rational coordinates independent of the Lambert-series identity. Put $a_n=A(n)/n$. For $N,D\in\mathbb{N}$, say that D clears the primitive Euler coordinates through N when $Da_n\in\mathbb{Z}$ for all $1\le n\le N$. Define

$$\lambda_N(p)=\begin{cases}4, & p=2 \text{ and } 4\le N,\\ p^2, & p \text{ is odd prime and } p^2\le N,\\ p, & p \text{ is odd prime and } p\le N<p^2,\\ 1, & \text{otherwise,}\end{cases}\qquad T_N=\operatorname*{lcm}_{p\le N}\lambda_N(p).$$

Proposition 2.3 (primitive-coordinate finite-jet obstruction)**.** *If $N\ge 4$ and D clears the primitive Euler coordinates through N, then $T_N\mid D$. Consequently no fixed positive integer clears all coordinates a_n.*

Formalised: `PriDetLif` and `PriDetLif`. The proposition concerns only these normalized finite coordinates. It makes no statement about arbitrary integral power-series jets, and it gives no irrationality criterion for S.

Finite Mersenne atoms. For a positive conductor d and tail index N, the pointed atom $2^{N \bmod d}/(2^d-1)$ obeys the exact doubling step with one carry at each wrap. For any finite matrix of such atoms, clearing the product of the column Mersenne denominators recovers an integer determinant. Thus a nonzero cleared minor has absolute value at least one; integer column weights do not create an additional height surplus. Formalised: `MerTaiAto`, `MerTaiAto`, and `MerTaiAto`. This is a finite-denominator boundary for an atom-minor route, not an irrationality criterion for S.

Proposition 2.4 (the Möbius-square lens)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \sum_{d\geq 1} \frac{\mu(d)}{(2^d-1)^2}.$$

Formalised: `CerKer`. Writing $T = \sum_{d\geq 1} \mu(d)/(2^d-1)^2$, irrationality of S is equivalent to irrationality of T, since $S = \frac{1}{2} + T$. The factor $1/(2^d-1)^2$ is the probability that d divides two independent fair-coin waiting times (Section 4.3). The signed constant, the coprime-pair mass, and the tail differences of Section 4.4 are three readings of the same quantity.

2.2 The squared transform and gcd moments

The square in Proposition 2.4 is not an isolated identity. For an arithmetic function f, put

$$L_2(f) = \sum_{d\geq 1} \frac{f(d)}{(2^d-1)^2}.$$

If X and Y are the independent waiting times used in Section 4.3, then

$$\Pr(d \mid \gcd(X,Y)) = Pr(d \mid X)\Pr(d \mid Y) = \frac{1}{(2^d-1)^2}.$$

Consequently $L_2(f)$ is the divisor calculus of the random gcd, just as $L(f)$ is the divisor calculus of one waiting time. More precisely, the formal transfer theorem gives, for $0 \leq r < 1$ and $|f(d)| \leq d$,

$$\sum_{d\geq 1} f(d)\left(\frac{r^d}{1-r^d}\right)^2 = \sum_{n\geq 1}\left(\sum_{e\mid n} f(e)\left(\frac{n}{e}-1\right)\right) r^n.$$

Formalised: `GcdMomCal`.

Proposition 2.5 (the squared Lambert ladder)**.** *At $r = 1/2$ the following identities hold:*

$$\begin{aligned}
L_2(\mu) &= S - \tfrac{1}{2},\\
L_2(\mathbf{1}) &= \sum_{n\geq 1} \frac{\sigma(n)-\tau(n)}{2^n},\\
L_2(\varphi) &= \sum_{n\geq 1} \frac{P(n)-n}{2^n}, \qquad P(n) = \sum_{e\mid n} \varphi(e)\frac{n}{e}.
\end{aligned}$$

Here P is Pillai's gcd-sum function. The three rows are respectively the coprime probability, the expected divisor count of $\gcd(X,Y)$*, and the expected value of* $\gcd(X,Y)$*.*

The first row is Proposition 2.4. The second and third are formalised at `GcdMomCal` and `GcdMomCal`. In the standard notation $\zeta_q(s) = \sum_{n\geq 1} \sigma_{s-1}(n) q^n$, the middle value is $\zeta_{1/2}(2) - \zeta_{1/2}(1)$. Postelmans–Van Assche prove that $1, \zeta_q(1), \zeta_q(2)$ are linearly independent over $\mathbb{Q}$ when $q^{-1} \in \mathbb{Z} \setminus \{-1, 1\}$ [13]; hence this middle value is irrational. Their Hermite–Padé argument is cited only and is not formalised here. It does not transfer irrationality to the Möbius row: $L_2(\mu) = S - 1/2$ remains precisely the open arithmetic atom.

The same pair calculus has a reduced-direction identity

$$\sum_{\substack{a,b\geq 1\\(a,b)=1}} \frac{1}{2^{a+b}-1} = 1,$$

formalised at GcdMomCal. Its Stern–Brocot cylinder mass $C(a,b) = ((2^a-1)(2^b-1))^{-1}$ satisfies an exact parent–children recursion, and the depth-d unfolding converges to $C(a,b)$ with error at most $(2/3)^d C(a,b)$; see GcdMomCal and GcdMomCal. These are exact representations of the same positive mass. They do not by themselves supply an irrationality criterion for S.

3 Erdős–Borwein-type irrationality (the #257 direction)

3.1 Full support: the Erdős–Borwein constant, every base

Theorem 3.1 (full-support irrationality)**.** *For every integer $b \geq 2$, the series $\sum_{n\geq 1} \frac{1}{b^n-1}$ is irrational. In particular the Erdős–Borwein constant $E = \sum_{n\geq 1} 1/(2^n-1)$ is irrational.*

Formalised: CerKer (all bases) and CerKer (the base-2 corollary). This is Erdős's 1948 theorem [1]. In the ladder of Section 2 it is the rung $L(\mathbf{1})$, and it is the fixed machine-checked point one convolution away from the open S.

The proof begins with the Lambert identity

$$\sum_{n\geq 1} \frac{1}{b^n-1} = \sum_{m\geq 1} \frac{\tau(m)}{b^m},$$

where $\tau(m)$ is the number of positive divisors of m (CerKer). If the right-hand side were rational with reduced denominator q, then a non-integral integer multiple could not lie at distance less than $1/q$ from an integer. It is therefore enough, for every q, to construct N such that $b^N \sum_m \tau(m) b^{-m}$ is non-integral but lies within $1/q$ of an integer. This elementary near-integer criterion is formalised at CerKer.

The construction of N is divided into three finite parts. First, a bounded Bertrand argument supplies disjoint prime blocks, and the Chinese remainder theorem chooses an arithmetic progression on which

$$b^r \mid \tau(N+r) \qquad (1 \leq r \leq K).$$

These divisibilities make the first K terms of the shifted tail integral. The multiplicativity step that converts prescribed prime valuations into the displayed divisibility is CerKer. Second, divisor pairing bounds the average of the weighted middle window

$$\sum_{K<r\leq L} \tau(N+r) b^{L-r}$$

along that progression. A pigeonhole choice then selects one translate with a small middle contribution; see CerKer. Third, the elementary estimate $\tau(n) \leq n$ controls the infinite tail after L. The parameters are chosen so that the middle and far-tail bounds together are smaller than $1/q$; the shifted tail is strictly positive because every divisor count is positive, so the resulting dilation is not an integer.

Thus the analytic series is consumed only at the Lambert identity and the geometric tail estimate. The remaining proof is a finite CRT construction, divisor counting, and explicit parameter arithmetic. The final certificate existence theorem holds for every $b \geq 2$ and every precision q (CerKer); its composition with the near-integer criterion is the theorem cited above. This is a formalisation of the classical full-support theorem, not a new irrationality result.

3.2 Named infinite-support cases

Beyond full support, the development proves irrationality for a family of infinite supports A, at every base $b \geq 2$. Each is a formalised case of the #257 statement family. None is the universal statement, and none is claimed as new mathematics.

Theorem 3.2 (support-class family)**.** *For every integer $b \geq 2$, the series $\sum_{n \in A} 1/(b^n - 1)$ is irrational for each of the following supports A:*

(a) *factorial support $A = \{n! : n \geq 1\}$, giving $\sum 1/(b^{n!} - 1)$;*

(b) *power-of-two support $A = \{2^n : n \geq 0\}$, giving $\sum 1/(b^{2^n} - 1)$;*

(c) *multiples $A = d\mathbb{N}$;*

(d) *pairwise-coprime supports with summable reciprocals (Erdős's condition [3]);*

(e) *eventually-periodic supports, residue classes, and the odd numbers.*

These sort into three groups. The classical case is (d): infinite pairwise-coprime A with $\sum_{a \in A} 1/a < \infty$, formalised from Erdős's condition. The lcm-gap named cases are (a) and (b), whose support gaps outgrow the running lcm; the structured families are (c) and (e).[1] Formalised: (a) `CerKer`; (b) `CerKer`; (c) `CerKer`; (d) `CerKer`; (e) `CerKer`, `CerKer`, `CerKer`. The two base-2 instances that name the problem directly are `CerKer` and `CerKer`.

The prime support is now known unconditionally: Tao and Teräväinen proved that

$$\sum_{p} \frac{1}{2^p - 1} = \sum_{n \geq 1} \frac{\omega(n)}{2^n}$$

is irrational [19]. Their proof uses quantitative two-point correlations of multiplicative functions and is not formalised here. This settles an important special case, while the universal support statement remains open.

The factorial and power-of-two cases are worth a word because they are not reachable by a Liouville shortcut: the power-of-two gap $2^k - \mathrm{lcm}$ stays within a bounded ratio of the lcm, so the terms do not decay fast enough for a transcendence-measure argument, and a uniform criterion is needed. The mechanism is *period noncollapse*: a prime-valuation-deficit witness forces the multiplicative order of the base modulo the relevant modulus not to collapse, which rules out the long repeated digit blocks a rational value would require.

3.3 The half-value problem in one page

The most concrete unresolved test value is $1/2$. The Mersenne weights are strictly superincreasing, so a normalized support representing a given value is unique. The greedy algorithm therefore does more than search for a support: it recovers the only possible one. For $1/2$, membership in the achievement set is equivalent to the greedy orbit skipping infinitely many exponents.

Suppose instead that the orbit has a last skip. At the following dyadic seam, the last contribution must come from either an upper cell or a middle cell. The upper alternative is impossible. Among the three exceptional middle coordinates $-3, -2, -1$, the coordinate -3 is also impossible: it would make the remaining cofinite tail smaller than $1/2$ while forcing a negative centred carry, contradicting a nonnegativity identity.

[1] Luca–Tachiya prove the broader eventual-periodic rational-coefficient theorem; its indicator specialisation directly contains case (e) [14]. Their Chowla–Erdős/ large-modulus-prime proof is distinct from the periodic-divisor certificate route formalised here, and the full mixed-sign coefficient theorem is not claimed here.

This leaves a precise open edge. It is enough to show that every remaining middle contribution is larger in magnitude than all of its future divisor-incidence contributions combined. That tail estimate is not proved for the actual greedy orbit. In particular, the formal source does not yet identify the two remaining coordinates $-2, -1$ with the failure of a dyadic safety condition. Thus the argument eliminates two branches and isolates the last one; it does not prove that $1/2$ belongs to the achievement set.

The exact statements are GreAchSet, GreAchSet, HalCylFinMidCelEsc, and HalCylLasProCon. The rest of this section develops sufficient criteria and finite case analyses around this remaining estimate; it may be skipped on a first reading.

3.4 A narrow obstruction for an isolated skip

One local configuration admits a particularly simple exact description. Suppose the greedy process takes the term at rank b, skips the next term, and then takes the term after that. Put $q = 2^b - 1$, and write the residual immediately before the first take as $1/R$. When $0 < R < q$, the intervening skip is dyadically unsafe if and only if

$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2}.$$

The width of this interval is exactly

$$\frac{q^2}{(3q+1)(3q+2)} < \frac{1}{9},$$

and the whole interval lies between $2q/3$ and $2q/3 + 2/9$. Thus an unsafe isolated skip is confined to a uniformly narrow window pinned just above two thirds of the Mersenne modulus. No integral reciprocal lies in this window. In the corresponding positive odd integer formulation, with residual $p/(2D)$, membership in the cleared band forces $p \geq 7$.

This is a localizer, not an orbit-level elimination. It does not show that the greedy orbit of $1/2$ avoids the interval, and it does not identify the two surviving last-producer coordinates $-2, -1$ with unsafe isolated skips. The exact equivalence, width calculation, and arithmetic consequences are HalGreTwoThiBan, HalGreTwoThiBan, HalGreTwoThiBan, HalGreTwoThiBan, and HalGreTwoThiBan.

3.5 A conditional orthogonal-petal criterion

Let A be a finite-core orthogonal-petal bouquet: outside a finite frame, its elements have the form $c_i p_i$, where the cores c_i divide one fixed integer and the nontrivial petals p_i are pairwise coprime to that frame and to one another, with summable reciprocal mass. The formal construction also contains an explicit infinite example with alternating cores 2 and 3 and squared prime petals; every member of this support has exactly three prime factors with multiplicity.

For such a bouquet, the remaining input is an exact tail-selection predicate: at every positive binary scale it chooses a carry slot with a normalized tail at most 16. Lean proves that this predicate supplies the needed forced-carry certificates and hence irrationality of the base-2 support series. The result is SupSunDic, with the carry bridge at SupSunDic. The tail-selection predicate itself is not proved here. Thus this is a conditional reduction for one structured support family, not a further unconditional case of Erdős #257.

3.6 Composite dilation defects

For a support divisor count $f_A(x) = \#\{d \in A : d \mid x\}$, multiplication by a composite support element a has an exact correction term. Besides the distinguished divisor a, the new divisors are counted by a finite defect which excludes divisors already present at x. The formal identity is `ComDilDef`. For a finite-core orthogonal-petal bouquet, the defect at a ray is bounded by the finite exceptional frame plus the divisor count of the petal support: `ComDilDef`. This records the local correction needed when dilation is not prime. It does not prove a correlation estimate, the sunflower tail selector, or an irrationality statement.

3.7 Mixed prime-power layers

The one-prime-power layer has a two-signature extension. Exact p-adic and q-adic layer operators commute, and, for distinct primes with the residual argument coprime to pq, their composite extracts the iterated exact-valuation pullback of the support coefficient. This is formalised by `MaxOmeLay`. The singleton support $\{12\}$ gives a checked $(2^2, 3)$ example with value one at residual argument 1: `MaxOmeLay`. These are finite coefficient identities. They do not provide a decimation transport, a bounded-Ω conclusion, or an irrationality theorem.

3.8 Half-trapping return carries

If rational-linear channels vanish on every relation killed by one scalar evaluation, their evaluation matrix has rank at most one; every nontrivial square minor therefore vanishes. The all-rank determinant statement is `HalTraRetCar`. For two reverse carries with a common coefficient word, a 1/0 seam, and a common following overlap, the terminal difference is a power of two times an odd integer. A shared nonnegative Archimedean bound then gives the matching dyadic spacing inequality, formalised at `HalTraRetCar`. The argument does not produce reverse-carry presentations or an earlier overlapping return; it supplies only the algebraic core of that conditional criterion.

3.9 Half-carry reachability

There is also a finite-prefix criterion for the target $1/2$. At an unresolved even seam, the scalar transition reaches every integer terminal carry except the single value $2\delta - c$; this is the exact equivalence `HalCarRea`. A canonical description of such a seam at every sufficiently large even depth is an unproved hypothesis. Under that hypothesis, the finite admissible words are cofinal, by `HalCarRea`, and compactness then produces an infinite support with sum exactly $1/2$: `HalCarRea`. Thus the formal result is a conditional reduction. It does not establish the canonical seam supply and does not prove Erdős #257.

Cofinal strip returns. The all-level strip hypothesis can be weakened for the canonical greedy half carry. Its uncentred, binary-scaled form is antitone. Consequently, if it returns to the square-root strip cofinally, then one sufficiently late return bounds every later scaled carry; the strip envelope divided by 2^N tends to zero. The resulting tempered centred carry gives an infinite greedy support with support-series value $1/2$.

Proposition 3.3 (cofinal strip-return closure)**.** *If the canonical greedy half carry returns cofinally to the stated square-root strip, then* $1/2$ *belongs to the Mersenne achievement set.*

Formalised: `CofStrRet` and `CofStrRet`. The cofinal return condition itself is not established. It is weaker than a uniform strip bound and still does not give an unconditional membership theorem.

Terminal-only cofinal approximation. Coherence can be removed entirely from a further version of this reduction. At depth M, take only a normalised finite word whose terminal half carry lies in the square-root strip. Its finite support then has support-series value within

$$\frac{4M+12}{2^M}$$

of $1/2$. If such terminal witnesses occur cofinally, these finite values converge to $1/2$; closedness of the Mersenne achievement set supplies the limit. No control of the earlier rows, and no compatibility between the witnesses, is used.

Proposition 3.4 (terminal-only cofinal closure)**.** *Cofinal normalised terminal witnesses in the square-root strip imply that* $1/2$ *belongs to the Mersenne achievement set, and hence that an infinite support has support-series value* $1/2$*.*

Formalised: `TerOnlCof`, `TerOnlCof`, and `TerOnlCof`. The cofinal terminal-witness supply is not proved. In particular, the proposition does not establish membership of $1/2$ unconditionally.

Scaled terminal vanishing. The square-root strip is stronger than the analytic condition used by the terminal-only argument. Let the depths of normalised finite words tend to infinity. If their terminal carries, after division by the corresponding power of 2, tend to zero, then the universal coefficient-tail error also tends to zero. The finite support values therefore converge to $1/2$, and closedness of the achievement set again supplies the limit. The earlier terminal-only strip condition implies this weaker sequential hypothesis.

Proposition 3.5 (scaled terminal-vanishing closure)**.** *A normalised terminal-word sequence with depths tending to infinity and vanishing binary-scaled terminal carries yields an infinite support with support-series value* $1/2$*.*

Formalised: `TerOnlScaVan`, `TerOnlScaVan`, and `TerOnlScaVan`. No such scaled-vanishing sequence is constructed here; the result is a conditional reduction, not an unconditional membership theorem.

3.10 Finite shadows of the half cylinder

For an admissible finite word, the residual from $1/2$ is given exactly by its terminal half carry minus the future coefficient tail, scaled by a power of two; see `HalCylFinSha`. At a first seam, avoiding the integer strip is equivalent to full scalar reachability of that strip: `HalCylFinSha`. The rank-three greedy example is raw dyadically safe while its seam has an in-strip hole, so raw safety alone does not imply this escape. Finally, an explicit frozen-margin producer implies $1/2$ belongs to the Mersenne achievement set, by `HalCylFinSha`. That producer is an unproved hypothesis; this is a conditional reduction, not a proof of Erdős #257.

3.11 Fixed-coefficient rewind

For a finite list of common coefficients, the canonical inverse-parent map has an exact closed form with a power-of-two denominator: `HalCarCeiParCon`. An interval of width at most that denominator contracts either to one ancestor or to two adjacent ancestors. The singleton alternative is equivalent to the explicit dyadic phase inequality, while the other alternative is a seam pair; see `HalCarCeiParCon`. Thus width alone does not justify a singleton conclusion. This is a finite rewind lemma, and supplies neither a seam exclusion nor an Erdős #257 conclusion.

3.12 Selected half-carry windows

The selected-window construction keeps an explicit admissible word for each terminal carry in a finite protected interval. A supply of such windows at all sufficiently large levels gives cofinal admissibility and hence an infinite support with sum $1/2$: `HalCarSelWin` and `HalCarSelWin`. The unresolved input is the coherent history at each step: either a common next-row divisor cylinder, or the canonical adjacent seam. The module makes this input explicit; it does not construct the window supply and does not prove Erdős #257.

3.13 Realized rewind phase

For a realized fixed-coefficient history, the dyadic rewind phase is the complementary binary suffix numeral: `HalCarRewPha`. Consequently the singleton criterion is exactly the inequality that this suffix numeral covers the rest of the terminal interval, formalised by `HalCarRewPha`. This converts the phase condition into support-history coordinates, but does not prove the required coverage inequality globally.

3.14 Localized protected seams

The compactness argument needs only one admissible word at cofinally many depths. At a localized one-hole seam, one of the two protected carries 3 and 4 survives; therefore a cofinal supply of realized protected seams implies cofinal admissibility, `HalCarProSeaCon`, and hence an infinite support with sum $1/2$, `HalCarProSeaCon`. The cofinal realization supply is not proved.

3.15 Combining the two cofinal half-carry criteria

At each requested scale, the public mixed producer permits either a selected protected window or a localized realized seam. Each branch supplies the one admissible word needed for compactness, so this cofinal disjunction implies an infinite support with sum $1/2$: `HalCarCofWinOrSeaCon` and `HalCarCofWinOrSeaCon`. The mixed cofinal producer remains an unproved hypothesis.

3.16 Integer-greedy first-wrap reduction

The first-wrap seam has exact truncated Mersenne weights with a quantitative gap-dominance property. For these weights, a small positive subset-sum defect exists exactly when the descending integer-greedy remainder is small and positive; this is formalised by `HalCylIntGre`. The remaining arithmetic input is an unbounded lower bound for that deterministic remainder sequence. No such lower bound is proved here.

3.17 Concrete seam adapter

The finite seam-word adapter realizes the abstract adjacent cut by concrete Boolean words and identifies its next word with the integer-greedy choice: `HalCylConSeaAda`. It implements a finite recurrence, not the unbounded remainder estimate.

3.18 Fatal gaps and eventual right tails

For the concrete integer seam words, eventual right extension is equivalent to nonmembership of $1/2$ in the Mersenne achievement set: `HalCylFatGapRigTai`. The same condition is equivalent to the existence of a finite fatal half gap, by `HalCylFatGapRigTai`, and also to a final skipped exponent in the real greedy orbit, by `HalCylFatGapRigTai`. Thus this equivalence gives an exact classification of the nonmembership branch. It neither decides whether $1/2$ belongs to the achievement set nor proves the universal Erdős #257 statement.

This is the nonmembership half of the argument summarized earlier: a last greedy skip is equivalent to eventual right extension. The following paragraph records the first exceptional middle case that can be ruled out.

First final-middle cell. The first exceptional middle cell has a separate obstruction: if the orbit extends right after the last false rank, then the cell with producer carry -3 makes the lazy cofinite tail strictly smaller than $1/2$ while its centred carry becomes negative. This is impossible by the analytic nonnegativity identity, `HalCylFinMidCelEsc`. It removes that one conditional terminal configuration; it does not decide the right-tail branch or the remaining two middle cells.

The remaining global reduction is now precise. If every genuine middle producer other than that discharged cell has producer carry larger than its complete future divisor-incidence tail, then an eventual right seam is impossible and $1/2$ belongs to the Mersenne achievement set: `HalCylLasProCon`. The stronger square-root lower-bound hypothesis is also sufficient, `HalCylLasProCon`. Neither hypothesis is supplied here.

For finite seam prefixes the remaining tail bound sharpens further: the future divisor-incidence tail is bounded by the cardinality of the prefix. Thus the explicit row-scale inequality $|c| + \mathrm{belowPulse} + 5 < 4\,\mathrm{remainder}$ is sufficient for the same half-membership conclusion, `HalCylMidCarLowBou`; it is enough in turn that every late middle remainder is at least its row, `HalCylMidCarLowBou`. These are exact conditional reductions, not proofs of the row bound.

Finite upper-reset band certificates. The companion upper branch has a kernel-checked finite certificate: for every upper transition from rows 13 through 30, every subsequent dyadic danger band is avoided. The certificate shares each exact successor remainder across all its band indices, `HalCylUppResBanCer`. It is a bounded producer check and does not supply the cofinal band-escape hypothesis.

3.19 Positive half-membership classification

The complementary branch has an equally exact seam formulation. Membership of $1/2$ in the Mersenne achievement set is equivalent to false successor terminal bits beyond every bound, `HalCylHalMemCla`; equivalently, it is equivalent to a cofinal sequence of such false terminals, `HalCylHalMemCla`. It is also equivalent to a cofinal integer-seam sequence with skipped ranks tending to infinity, `HalCylHalMemCla`. These equivalences isolate the positive arithmetic producer; they do not establish that the producer exists.

3.20 Largest skipped ranks

At a named largest skipped rank d, the lower seam word and its adjacent upper competitor have an exact late-range gap. In division-free form, the identity is

$$3L + \left(3 \cdot 2^{s+1} + 2 \cdot 4^{s-d} + 4\right) = 3U$$

whenever $2s < 3d$; this is formalised at `HalCylLarSkiGap`. The upper and middle successor branches create a terminal largest skipped rank, while the right branch preserves an existing one: `HalCylLarSkiGap` and `HalCylLarSkiGap`. The first-crossing branch producer required to use this identity globally is not proved.

3.21 Boundary-pulse normalization

Above a late largest skipped rank, every filled suffix rank has zero row pulse, so the word pulse is carried entirely by the lower prefix: `HalCylBouPul`. At the first crossing of the two-thirds boundary there are exactly two cases, $3d = 2s + 1$ and $3d = 2s + 2$, and the skipped rank has pulse respectively two or one: `HalCylBouPul`. This supplies the incidence normalization for a producer estimate; it does not establish that estimate.

3.22 Largest-skip induction

Assume that at the first crossing a late largest skipped rank either remains late at the next row or forces an upper-or-middle successor branch. This hypothesis propagates a late largest skipped rank from row 14 onward, `HalCylLarSkiInd`, and hence yields cofinal skipped ranks and membership of $1/2$ in the Mersenne achievement set, HalCylLarSkiInd. The first-crossing hypothesis itself remains unproved.

3.23 The fixed-tail survival criterion

A skipped rank is final exactly when its post-decision remainder lies strictly above the complete remaining Mersenne tail: HalCylFixTaiSoc. Consequently membership of $1/2$ is equivalent to the pointwise assertion that every actual skipped rank has remainder at most its full future tail, HalCylFixTaiSoc. This is an exact local reformulation; the tail-survival inequality is not proved.

3.24 The producer-carry identity

For a producer-aligned finite support, the residual from $1/2$ is exactly the integer producer carry minus the complete future divisor-incidence tail, scaled by $2^{-(2d+2)}$: HalCylProCarSoc. Thus the carry exceeds that tail exactly when the support series lies below $1/2$, and a negative carry places it above $1/2$; HalCylProCarSoc and HalCylProCarSoc. An upper carry at most -8 even gives a full Mersenne-gap margin, HalCylProCarSoc. These are local criteria, not a proof that the required carries occur.

3.25 Quarter-band endpoint cells

For the adjacent seam recurrence, failure of the desired next-row upper bound on the middle branch is exactly one multiple-of-four pulse cell, HalCylQuaBanEnd. On the terminal-reset branch, the same failure is exactly a constrained $3 \cdot 2^{s-1} + k$ pulse cell, HalCylQuaBanEnd. These are arithmetic normal forms for the exceptional cells; no avoidance statement for the concrete seam orbit is proved.

3.26 Reset-deficit escape

At a late largest skipped rank, a right successor branch pins the integer greedy remainder between an explicit quarter threshold and carry threshold: HalCylResDefEsc. Along such a branch the remainder satisfies an exact affine recurrence, HalCylResDefEsc. Moreover, a doubled-scale deficit forces an upper-or-middle branch within the corresponding number of rows, HalCylResDefEsc. The remaining input is the deterministic anti-concentration needed to exclude the pinned right-branch window globally.

3.27 Seam-to-producer carry alignment

At a first-wrap seam, the producer-aligned carry of the left boundary support is four times the shifted seam hole, less the explicit paired incidence pulse at the next two rows: HalCylSeaProAli. This is an exact coordinate change between the seam and producer pictures; it does not supply a sign or size estimate for that carry.

3.28 A computed selected-window base

The selected half-carry construction has a checked depth-18 base: twelve Boolean words cover every terminal carry in the strip $[1, 12]$, with all intermediate carries in the corresponding half strip. The reflected table is formalised at HalCarSelWinBas; the resulting selected window is defined at HalCarSelWinBas. All twelve words share the same restriction at depth 13, giving the first next-row divisor-agreement certificate, HalCarSelWinBas. This finite base does not provide selected windows at unbounded depths.

3.29 Operational rewind-seam bridge

For a selected window with a rewind seam, a two-value coefficient profile on the adjacent base ancestors induces the live next-row profile, `RewSeaOpeBri`. Under the stated strip and buffer inequalities, that profile realizes a protected one-hole seam at the next row, `RewSeaOpeBri`. The base-ancestor coefficient drop is an explicit input, not a theorem here.

3.30 Selected suffix cylinders

The depth-18 selected window has one common terminally indexed suffix cylinder after its depth-13 prefix is removed, `SelSufCyl`. Under the canonical selected-window successor, the endpoint evolves exactly by $E' = 2E - C$ and still covers the new carry interval, `SelSufCyl`. This preserves a finite cylinder invariant; it does not produce the required coefficient profile at every later step. For a rewind seam, the next prefix bit has an exact exhaustive outcome: the cylinder promotes by one bit, or the two base ancestors form the literal half-divisor boundary pair and hence produce a protected seam, `SufCylPro`. At an exact half-divisor row, the crossing-cylinder profile gives the more concrete advance-or-seam alternative, `SufCylGloPro`. The below-strip and above-strip hole cases have explicit finite word and suffix-cylinder sheet witnesses, `SufCylBel` and `SufCylTwoShe`; combining them leaves only a realized seam whose exact hole is inside the next strip, `SufCylFeeFanIn`. The strengthened in-strip alternative retains its two endpoint cylinders, complementary literal prefixes, every admissible word off the hole, and the canonical parent-failure coordinate required by the finite-gap construction, `SufCylInStr`. The one-hole state embeds as a singleton finite gap; provided a common next-row coefficient and the stated child-gap and buffer inequalities, this two-sheet package advances one row while preserving its complete data, `SufCylFinGap`. After cutoff promotion the two live sheets may have different coefficients. The profiled-gap interface retains their independent literal prefixes, derives the ready next-row profile, and advances under its explicit gap and buffer hypotheses, `SufCylProGap`. At the next cutoff boundary, the lower and upper sheets respectively gain their forced false and true prefix bits; the upper endpoint is adjusted by the corresponding head weight, `SufCylProPro`. The two retained prefixes remain adjacent, so their next-row coefficients differ by at most one in the directed form needed for the profiled update, `SufCylProAdj`. The same adjacency invariant survives the false/true cutoff promotion, `SufCylProProAdj`. Given its explicit next-row arithmetic cell, the promoted stage then advances with its coefficient profile and buffer synthesized internally, preserving adjacency through the whole feedback operation, `SufCylProAutSte`. If the resulting macro-step swallows the next strip, its coefficient-pulse bound forces the preceding gap into the exact quarter-band lower bound; this is a quantified residual branch, not an impossibility claim, `SufCylProSwa`. An explicit full-cylinder run reaches row 51 with endpoint 51,327,745, which exceeds the certified feedback head threshold and promotes the next cutoff, `SufCylSta`. Any cofinal family of full suffix-cylinder stages would yield the terminal-only cofinal producer consumed by the final support construction, `SufCylTerOnlBri`. This is explicitly conditional: the public development does not establish such a cofinal family. This is a conditional local feedback classification, not a cofinal supply theorem: the required crossing coefficient profile is still an input.

3.31 Pre-feedback selected-cylinder run

Before the first half-divisor feedback row, Lean constructs a full selected window at every depth from 18 through 27, `SelSufCylPreFee`. At depth 27 the common suffix cylinder remains and its endpoint covers the whole protected strip, `SelSufCylPreFee` and `SelSufCylPreFee`. The shared depth-13 prefix gives an explicit checked coefficient recurrence, including the resulting depth-27 endpoint, `SufCylFirFee`. This is a finite run, not an unbounded induction. The threshold-stage continuation extends this explicit window through depth 29: its endpoint still covers the strip and its common suffix cylinder is certified at every cutoff from 14 through 25, while cutoff 26 is explicitly not common, `SufCylThr` and `SufCylThr`.

3.32 Half-divisor unit drop

At the doubled feedback row, changing only the terminal half-divisor bit from false to true increases the support coefficient by exactly one: `HalDivUniDro`. The same holds for any explicitly identified adjacent boundary pair, `HalDivUniDro`. The rewind adapter transfers such a pair to the next protected seam while retaining the exact unit-drop witness, `RewHalDivAda` and `RewHalDivAda`. This is the local coefficient input for the rewind-seam profile, not a global construction of such boundary pairs.

3.33 A finite rewind-boundary obstruction

The scalar one-row rewind data alone still does not supply that missing boundary predicate. Lean gives a concrete depth-26 selected two-word window whose depth-27 explicit step has rewind history [1], is a width-two seam pair, and satisfies the doubled-row identity, but has no depth-13 half-divisor boundary-pair witness: `RewBouPaiCou`. Thus the protected-window buffer in the cofinal consumer is a genuine hypothesis, not a consequence of the scalar seam facts alone.

3.34 Concrete producer-carry coordinates

For every adjacent cut at rank $s \geq 5$, Lean identifies the producer carry of the terminal-augmented below support with the concrete middle coordinate:

$$4\,\mathrm{remainder} - \mathrm{belowPulse} - 4,$$

and identifies the terminal-augmented above support with the negative upper coordinate

$$-\big(4\,\mathrm{overshoot} + \mathrm{abovePulse} + 4\big).$$

The generic row-word identity is `HalCylProLowBou`, with the below and above specializations at `HalCylProLowBou` and `HalCylProLowBou`. These identities make the remaining quantitative task exact: they do not prove exponential lower bounds, sign persistence, or a universal escape mechanism.

3.35 What stays open

The universal Erdős #257 statement, irrationality of $\sum_{n\in A} 1/(2^n-1)$ for *every* infinite A, is not proved. The proved cases are a family, not the quantified theorem.

4 The totient constant S (Erdős #249)

Whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational is open. The development gives two things and claims neither as a solution: an unconditional exclusion of small-denominator rationals, and a conditional reduction to finite certificates.

4.1 The certificate argument in one page

Write

$$R_N = \sum_{m\geq 1} \frac{\varphi(N+m)}{2^m}.$$

For every positive shift h, integrality of even one difference $R_{N+h} - R_N$ forces S to be rational. Consequently S is irrational if and only if all these differences are non-integral. This is an exact equivalence, not merely a one-way reduction.

Non-integrality at fixed h and N has a complete finite test. Truncate the binary expansion at depth L, compute the resulting integer discrepancy, and check that its residue modulo 2^L lies farther from both endpoints than the rigorously bounded omitted tail. Such a certificate exists for some L if and only if $R_{N+h} - R_N$ is non-integral. Thus the infinite statement has an exact finite witness at each fixed pair of parameters.

There is also a direct analytic way to force one of these witnesses. On a finite dyadic block, attach to each discrepancy residue its first cosine harmonic. Under the explicit room inequality, any fixed saving below the near-integer value forces at least one residue into the central certified region. What remains is to prove such a harmonic saving at cofinally many parameters. The formal development proves the finite implication, but no cofinal estimate and hence no irrationality theorem for S.

The exact equivalences and finite implication are `LcmConFla`, `LcmConFla`, and `FirHarGap`. The remainder of this section gives the denominator theorem, the geometry of S, and the finite arithmetic behind this summary.

4.2 Unconditional: no small denominator

Theorem 4.1 (denominator exclusion)**.** *Set*

$$Q_0 := 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}.$$

For every integer p and every q with $1 \le q \le Q_0$,

$$S \neq \frac{p}{q}.$$

Equivalently, if S is rational then its reduced denominator exceeds that bound.

Formalised: `CerKer`. The bound is reached by a ladder of finite, elementary exclusions,

$$4838 \;\to\; 4\,194\,304 = 2^{22} \;\to\; 2.49 \times 10^{17} \text{ (Farey } K{=}120) \;\to\; 7.96 \times 10^{34} \text{ (Farey } K{=}240),$$

each a mediant/Farey argument that traps any candidate m/q inside a forbidden gap. The mediant framework is historically associated with Farey's 1816 note on vulgar fractions [11]; the finite exclusions themselves are Lean-checked arguments in the formal-source checkpoint. See `CerKer` for the first rung and `GapFarBou`, `GapFarBou` for the two Farey windows.

This is an explicit lower bound on the denominator of any rational representation, not a proof of irrationality. A rational with a larger denominator is not excluded by it.

Remark (one bound, four representations)*.* Because the ladder identities of Section 2 are equalities, the same finite record transfers verbatim to three other representations of the constant: to the positive lift $\sum_d A(d)/(2^d - 1)$ (`CerKer`), to the signed Möbius-square constant $T = \sum_d \mu(d)/(2^d - 1)^2$ at half the bound (`CerKer`), and to the coprimality probability of Section 4.3 at half the bound. Half appears because $S = \frac{1}{2} + T$ turns a denominator d for T into $2d$ for S.

4.3 The geometric picture: S as coprime-pair mass

Let X, Y be independent fair-coin waiting times, $\Pr(X = n) = 2^{-n}$ for $n \ge 1$. Then $\Pr(d \mid X) = 1/(2^d - 1)$, so the Lambert transform is a divisor calculus of X: $L(f) = \mathbb{E}[(f * \zeta)(X)]$. The rungs read as $L(\mu) = \Pr(X = 1) = 1/2$, $L(\varphi) = \mathbb{E}[X] = 2$, $L(\mathbf{1}) = \mathbb{E}[\tau(X)] = E$, and $L(A) = \mathbb{E}[\varphi(X)] = S$.

Counting visible lattice points (a, b) with $a + b = n$, $a > 0$, $\gcd(a, b) = 1$ gives exactly $\varphi(n)$, uniformly in n, so S is itself a coprime-pair mass.

Proposition 4.2 (coprimality probability)**.**

$$S \;=\; \tfrac{1}{2} \;+\; \Pr\bigl(\gcd(X,Y)=1\bigr) \;=\; \tfrac{1}{2} \;+\; \sum_{\substack{a,b\geq 1\\ \gcd(a,b)=1}} 2^{-(a+b)}.$$

Formalised: `CerKer`, with the underlying lattice identity at `CerKer`. Base 2 is the one point where $\Pr(X=a)\Pr(Y=b)=2^{-(a+b)}$ needs no normalising constant, so the totient sum and the coprimality probability line up with no boundary correction. In this reading, Erdős #249 asks whether two independent fair-coin waiting times are coprime with irrational probability.

4.4 Conditional: reduction to finite certificates

The reduction uses only the elementary bound $\varphi(n)\leq n$, geometric tail estimates, and finite integer arithmetic. For $t\geq 1$, write $M_t=\operatorname{lcm}(1,\dots,t)$.

Step 1: rationality forces an integer tail-period law. Write the fractional layer of $2^N S$ as the tail

$$R_N \;=\; \sum_{m\geq 1} \frac{\varphi(N+m)}{2^m}.$$

If S is rational, its binary expansion is eventually periodic, which forces, for a suitable period h and all large N, the integrality $R_{N+h}-R_N\in\mathbb{Z}$. To refute integrality at a finite depth L, form the window discrepancy

$$\Delta_{h,N,L} = \sum_{j=0}^{L-1} \bigl(\varphi(N+h+1+j)-\varphi(N+1+j)\bigr)\,2^{L-1-j}\in\mathbb{Z},$$

the first L binary digits of $R_{N+h}-R_N$ up to a controlled deep tail (bounded using only $\varphi(n)\leq n$), and call the pair *certified* when the residue of $\Delta_{h,N,L}$ modulo 2^L misses a neighbourhood of 0:

$$\mathsf{certifiedKill}(h,N,L):\quad N+h+L+2 \;<\; \bigl(\Delta_{h,N,L} \bmod 2^L\bigr) \;<\; 2^L-(N+h+L+2).$$

This is a finite integer computation, decidable by the kernel. It is the natural finite object here because a rational S would pin the tail difference to an integer, and a residue that stays a fixed distance from 0 is a finite proof that it did not.

Corollary 4.3 (exact tail-period normal form)**.** *S is irrational if and only if, for every period $h\geq 1$ and every threshold N_0, there exist $N\geq N_0$ and L with $\mathsf{certifiedKill}(h,N,L)$.*

Formalised: `LcmConFla`; the tail R_N and the predicate are `TotTaiPerKil` and `TotTaiPerKil`.

Step 2: collapse the two parameters to one. Every multiple of a period is a period, so $\operatorname{lcm}(1,\dots,t)$ is the universal-period envelope: any eventual period of a rational S divides it once t is large. Standing on that one diagonal ray covers all possible periods at once, and the position parameter drops out too.

Corollary 4.4 (diagonal reduction)**.** *Suppose that for every threshold t_0 there are parameters $t\geq t_0$ and L for which $\mathsf{certifiedKill}(M_t,M_t,L)$ holds. Then S is irrational.*

Formalised: `LcmDiaRed`.

Step 3: the cone, flatness, and completeness. On the cone $\{kM_t : k \geq 1\}$, rationality forces a single fractional constant across the whole cone.

Proposition 4.5 (cone flatness)**.** *If S is rational, then for every sufficiently large t there is $\theta_t \in [0,1)$ such that the fractional part of the tail R_{kM_t} equals θ_t for every $k \geq 1$.*

Formalised: `LcmConFla`.

The certificates are exact: a certificate exists exactly when the object it certifies is non-integral. Thus the finite computations are machine-checked witnesses rather than numerical experiments.

Proposition 4.6 (certificate completeness)**.** *For all h, N,*

$$\big(\exists L,\ \mathit{certifiedKill}(h,N,L)\big) \quad \Longleftrightarrow \quad R_{N+h} - R_N \notin \mathbb{Z}.$$

Formalised: `LcmConFla`. Conversely, integrality of one positive-shift tail difference forces S to be rational. Thus S is irrational if and only if every positive-shift tail difference is non-integral, equivalently if every such pair admits a finite certificate. These normal forms are formalised at `LcmConFla`, `LcmConFla`, and `LcmConFla`. Finite certificates witness non-integrality. A joint refuter can interrogate a menu of cone vertices at once and is sharper than any pairwise test: it refutes the single-fractional-part model that rationality predicts on a finite sample of cone vertices (`LcmConNon`).

A reduced-denominator finite predicate. Separately, the formal development records a stricter finite condition for an odd reduced denominator u. Its predicate $\mathsf{ReducedDenominatorUnitGapCert}(u,N,K)$ asks that every candidate integer in the associated dyadic tail interval be non-coprime to u. At a prime power $u = p^e$, the condition is equivalent to the exact alternative that the interval has no candidate, or that it has one candidate and p divides that candidate. Thus such an interval has at most one candidate.

Proposition 4.7 (prime-power unit-gap ceiling)**.** *For every prime power p^e with $e > 0$, a $\mathit{ReducedDenominatorUnitGapCert}(p^e,N,K)$ has candidate count at most one. The row $(u,N,K) = (3,3,5)$ satisfies this predicate, although its corresponding ordinary empty-gap inequality fails.*

Formalised: `PriWeiCer`, `PriWeiCer`, `PriWeiCer`, and `PriWeiCer`. This is a finite certificate-level strengthening. We do not establish that rationality of S supplies these predicates at unbounded parameters.

The certificates are non-empty. The initial segment of the required supply is machine-checked, so the reduction is not vacuous. By completeness, each of these is a kernel-checked proof of a specific non-integral tail difference, not a numerical experiment.

Example 4.8 (verified finite instances)**.** The following are checked by the Lean kernel:

- $\mathsf{certifiedKill}(h,12,16)$ for every period $h \in [1,8]$;
- certified kills for every period $h \in [1,16]$ at a fixed window;
- diagonal certificates $\mathsf{certifiedKill}(M_t,M_t,L)$ at the 28 scales $t \in \{1,2,3,4,5,7,8,9,11,13,16,17,19,23,25,27,29,31,32,37,41,43,47,49,53,59,61,64\}$;
- a tabulated joint cone-vertex certificate.

Formalised: `TotTaiPerKil`, `CarSurExt`, `LcmDiaRed` (with `LcmDiaRed`, `LcmDiaRed`), `DiaPinCerT64`, and `LcmConNon`. These instances show that each layer of the reduction is non-vacuous and, by completeness, exact. They are not evidence that the unbounded supply exists.

The first row of this finite table can be read without any search procedure. Take $(h, N, L) = (1, 12, 16)$. The integer in the definition is

$$\Delta_{1,12,16} = \sum_{j=0}^{15} \big(\varphi(14+j) - \varphi(13+j)\big) 2^{15-j} = -143140.$$

Its least nonnegative residue modulo 2^{16} is 53468. The tail allowance is $N + h + L + 2 = 31$, and hence

$$31 < 53468 < 65536 - 31 = 65505.$$

This is precisely $\mathsf{certifiedKill}(1, 12, 16)$. The soundness theorem (`TotTaiPerKil`) therefore gives $R_{13} - R_{12} \notin \mathbb{Z}$. It is one exact non-integral tail difference, not an assertion that such differences occur at unbounded scales.

An exact target interval. The diagonal calculation also admits a more local normal form. Put $H_t = M_t$ and

$$D_t = R_{2H_t} - R_{H_t}.$$

The formal development separates an exact rational Möbius shadow $Q_t = a_t/d_t$ in lowest terms and defines the remaining *foreign defect* by $F_t = D_t - Q_t$. The full target is the residue class of $-a_t$ modulo d_t:

$$\mathsf{FullTargetHit}(t) \quad \Longleftrightarrow \quad \exists k \in \mathbb{Z}, \qquad d_t F_t = -a_t + d_t k.$$

Since $D_t = a_t/d_t + F_t$, this is exactly the condition $D_t \in \mathbb{Z}$; no complementary term is discarded or assigned a conjectural sign.

Proposition 4.9 (diagonal pincer decomposition)**.** *For every t, diagonal integrality is equivalent to the foreign diagonal defect meeting the full target. Consequently, full-target avoidance at unbounded scales implies irrationality of S.*

Formalised: `DiaPinDec` and `DiaPinDec`. The proposition is an exact reduction; the unbounded avoidance hypothesis is still open.

Two algebraic inputs make this interval test explicit. First, truncate the Möbius-square identity of Proposition 2.4 after D terms. The resulting rational quantity gives a Lambert-projected centre $C_{H,D}$ for the scaled diagonal difference. The error is exactly a signed squared-Mersenne tail:

$$\mathsf{scaledDiagonal}(H) - C_{H,D} = c_H \sum_{d>D} \frac{\mu(d)}{(2^d-1)^2},$$

where c_H is the explicit diagonal coefficient in the formal source. The absolute value of the sum is bounded geometrically by $4/[3(2^{D+1}-1)^2]$. Thus a rational separation of the centre from every integer by more than the scaled bound proves target avoidance; the argument does not estimate the sign of the tail.

Proposition 4.10 (squared-Mersenne enclosure)**.** *The scaled diagonal tail difference differs from its Lambert-projected centre by the squared-Mersenne tail. Separation of the centre from the full target by more than the proved tail bound forces target avoidance.*

Formalised: `SquMerDiaEnc` and `SquMerDiaEnc`.

Finite foreign-residue projection. There is a second finite presentation of the same obstruction. For a cutoff D, split the finite Möbius residue diagonal into the channels $d \le D$ which divide H and the remaining channels. The divisor part is exactly the truncated Möbius shadow, while every retained nondivisor channel is explicit. Above the stable cutoff $2H$, a finite window of omitted channels is bounded by

$$c_H \left(\frac{2}{2^D} + \frac{4}{3\,4^D} \right),$$

where c_H is the diagonal coefficient. Thus, if the actual foreign defect agrees with this projection within the displayed bound, and the projected quantity is farther than that bound from every integer, then the full target is missed.

Proposition 4.11 (finite foreign-residue projection)**.** *The finite residue diagonal splits exactly into projected foreign and divisor channels. A controlled projected foreign defect with the stated integer separation cannot meet the full target.*

Formalised: `ActForResPro`, `ActForResPro`, and `ActForResPro`. The passage from finite windows to the analytic foreign complement, including the corresponding kernel identity, is not supplied here. Nor is an unbounded family of separated projections; these are parts of the open certificate-supply problem.

Second, the rational Möbius shadow has an exact reduced denominator. If r_t is the square-free kernel of H_t, $s_t = H_t/r_t$, and $J(r_t)$ is the odd Jordan scalar defined in the algebraic shadow chain, then

$$d_t = \frac{2^{r_t} - 1}{\gcd\bigl(2^{r_t} - 1,\, s_t |J(r_t)|\bigr)}.$$

For $t \ge 5$, the product of the Mersenne factors $2^p - 1$ over primes $t/2 < p \le t$ divides d_t; in particular $d_t \ge 2^{t/2}$. This is a denominator theorem for Q_t only. Cancellation by F_t is exactly what the full-target condition retains.

Proposition 4.12 (Mersenne-shadow denominator)**.** *At lcm height t, the reduced denominator of the scaled Möbius shadow is given exactly by the formal denominator formula. The upper-half Mersenne product divides it and yields the stated lower bounds.*

Formalised: `MerShaDenGro` and `MerShaDenGro`.

Finally, the remaining fresh-loss term can be represented by a finite integer projection, and then by binary suffix data. At depth L, define

$$W_{t,L} = P_L(2H_t) - P_L(H_t), \qquad \rho_{t,L} = W_{t,L} \bmod 2^L,$$

where $P_L(M)$ is the length-L window numerator from Step 1. The finite residue certificate is the asymmetric safe interval

$$H_t + L + 2 < \rho_{t,L} < 2^L - (2H_t + L + 2).$$

The two unequal margins are the tail bounds at the two cone vertices H_t and $2H_t$; replacing them by a symmetric slogan would lose part of the checked statement.

Corollary 4.13 (fresh-loss projection reduction)**.** *The fresh-loss residue certificate is equivalent to its projection inequality, and the projection forces full-target avoidance. Hence an unbounded projection supply, or either stronger adjacent-suffix supply stated in the formal development, implies irrationality of S.*

Formalised: `DiaFreLosBri` and `DiaFreLosBri`. The development proves finite fixtures for this bridge but does not assert an unbounded projection or suffix supply.

T64 certificate closure. The generated Lucas-certificate closure supplies the primality dependencies for the finite diagonal-pincer certificate family through $t = 64$; for example, `DiaPinPriCerT64`. It is finite certificate infrastructure, not an unbounded certificate supply or an irrationality proof.

T64 endpoint certificate. Using that closure, Lean checks the individual diagonal-pincer certified kill at $t = 64$, `DiaPinCerT64End`. This is one finite endpoint certificate, not an unbounded supply.

Flexible odd three-rank window. At an analytically admissible odd power-of-two rank, centrality at any one of the canonical odd rank or its next two base-four successors yields the landed adjacent-suffix gap interface, `DiaFleOddWinSup`. Thus a cofinal three-rank band supply implies irrationality, `DiaFleOddWinSup`; that supply is not proved.

Affine frontier for that window. With the stated short-correction bounds, the three central-band predicates are equivalent to one three-scale affine escape condition, `DiaFleOddWinAff`. A zero-start affine orbit shows that a uniform correction envelope alone does not force escape at any finite number of ranks, `DiaFleOddWinAff`.

Fresh-prime deficit decomposition. The literal foreign channel also has an endpoint form suited to curvature tests. Retain only the prime support already present at H in the two endpoints $H + s$ and $2H + s$, and write the resulting discrepancy from the actual totient as an endpoint deficit. Each such deficit is nonnegative. The actual diagonal increment then equals the old-prime increment plus the lower deficit minus the upper deficit. The second and five-point curvature operators preserve this exact split; the latter has coefficient row $[-8, 4, 2, 1, 1]$. Consequently, the old curvature margin loses at most the stated adverse weighted deficit.

Proposition 4.14 (fresh-prime deficit decomposition)**.** *The endpoint fresh deficits are nonnegative, give an exact decomposition of the foreign channel, and bound the loss from the old-prime curvature margins.*

Formalised: `FrePriDefDec`, `FrePriDefDec`, and `FrePriDefDec`. This supplies the exact correction term for the finite probes. It supplies neither a cofinal deficit bound nor an irrationality criterion.

One-bit lifting. The power-of-two signed-margin socket admits a separate exact arithmetic reduction. If a signed difference is already divisible by 2^k, lifting its congruence to 2^{k+1} is equivalent to evenness of the quotient after division by 2^k. Failure of the lift gives the unique translated residue class. In particular, a displayed difference $P - N = 16c$ agrees modulo 32 exactly when c is even; otherwise the residue is the offset class by 16.

Proposition 4.15 (one-bit signed-margin lift)**.** *The fifth-bit question after fourth-bit agreement is exactly one cofactor parity test, with the complementary offset class characterising failure.*

Formalised: `PowTwoBitLif`, `PowTwoBitLif`, and `PowTwoBitLif`. The centered-state formulation identifies the signed-margin socket with failure of this next lift, subject to its stated correction and margin budget: `PowTwoCenBitLif` and `PowTwoCenBitLif`. This is a finite arithmetic interface. It does not supply the required cofactors at unbounded scales and gives no irrationality conclusion.

Parity is not a margin certificate. The preceding lift does not settle the reduced signed-margin socket. For every positive even modulus and every radius in the stated nontrivial range, each parity class contains both a socket hit and a socket miss. The same statement applies to a positive power-of-two modulus. Thus even a fifth-bit computation, by itself, cannot decide the required margin condition.

Proposition 4.16 (parity obstruction for the reduced margin socket)**.** *Under the explicit radius inequalities, parity does not decide membership in the reduced margin socket, including at a positive power-of-two modulus.*

Formalised: `TotParSocCei` and `TotParSocCei`. The module contains no $t = 827$ numeral and proves no signed-margin certificate; it excludes parity as a sufficient standalone criterion.

A closed prime-adjunction route. One tempting way to seek an unbounded obstruction is to compare the full target at H, pH, qH, and pqH. The formal development rules out this route in its unprojected form. For every positive k, the actual diagonal satisfies an affine transport law

$$D_{kH} = Q_k(H)D_H + Z_k(H), \qquad Q_k(H) \in \mathbb{N}, \quad Z_k(H) \in \mathbb{Z}.$$

Hence integrality at H propagates to every multiple, and the four target hits at the prime-adjunction diamond are equivalent to the single hit at its root (`FulTarPriAdjNoGo`). The corresponding transport of the complete foreign defect has zero diamond curvature (`FulTarPriAdjNoGo`). Thus explicit-shadow fibres alone cannot create a holonomy obstruction. A future positive route would need a separately defined projection together with a theorem controlling what that projection discards. This is a proved obstruction to one strategy, not progress on the unbounded-supply proposition.

Proposition 4.17 (scalar-localisation obstruction)**.** *Let $x \in \mathbb{Q}$ and $c \in \mathbb{Z}$. If $H \mid \mathrm{den}(x)$ and the reduced denominator of cx divides H, then*

$$\frac{\mathrm{den}(x)}{H} \mid |c|.$$

Formalised: `AdeHeiObs` and `AdeHeiObs`. Thus a scalar localisation can move a complementary denominator factor into its coefficient, but cannot erase it. This rules out a scalar-denominator shortcut of that form. It neither proves full-target avoidance nor supplies any certificates, so it gives no irrationality criterion for S.

Proposition 4.18 (square-CRT correction suppression)**.** *Let E be a finite family indexed by distinct primes p_i, with prescribed anchors A_i and residues a_i. There is a bounded common base n such that, for each $i \in E$ and each shift h with $p_i \nmid a_i + h$, one has*

$$n = A_i + p_i(a_i + p_i t) \quad \text{and} \quad \varphi(n + p_i h - A_i) = (p_i - 1)\varphi(a_i + p_i t + h)$$

for a suitable $t \in \mathbb{N}$. In particular, if every p_i exceeds a fixed horizon J, the choice $a_i = 0$ works simultaneously for all $1 \le h \le J$.

Formalised: `SquCRTCub` and `SquCRTCub`. The finite mechanism does not force a cube coefficient to be nonzero: there is a checked clean two-step cube with both displayed coefficients zero (`SquCRTCub`) and a separate checked clean cube with second coefficient -4 (`SquCRTCub`). Thus the result gives neither an unbounded certificate supply nor an irrationality criterion for S.

Proposition 4.19 (signed dyadic-moment algebra)**.** *For matrices $M : \iota \times \kappa \to R$ and $N : \kappa \times \iota \to R$ over a commutative ring, the determinant of MN is the sum over all maps $p : \iota \to \kappa$ of the determinant of the corresponding selected-column matrix times the diagonal product $\prod_i N_{p(i),i}$. Separately, if a selected integer numerator is odd and its dyadic exponent is strictly larger than every other selected exponent, then the common numerator after dyadic denominator clearing is odd and hence nonzero.*

Formalised: `SigQMomObs`, `SigQMomObs`, and `SigQMomObs`. These are finite algebraic inputs to a signed-moment route. They do not show that any actual totient Hankel determinant is nonzero, and they give no irrationality criterion for S.

Proposition 4.20 (dyadic totient certificate interface)**.** *For every fixed level e, a separated nonzero evaluation minor of the canonical dyadic totient-kernel family gives its span dimension 2^e+1. Every even residue channel is exactly a scalar multiple of its odd-core channel. Moreover, a compressed-adjoint certificate whose nonzero multiple is strictly smaller in absolute value than its positive modulus is inconsistent.*

Formalised: `TotMahDef`, `TotMahDef`, and `TotMahDef`. The unformalised producer is the all-level construction of the separated minors (by a CRT–Dirichlet argument); the interface does not provide those certificates for every e, and it gives no irrationality criterion for S.

Proposition 4.21 (residual-gauge obstruction for monomial minors)**.** *Let a monomial matrix have a common nonzero residual weight in each column. The residual weights act by a right diagonal gauge, and therefore preserve determinant nonvanishing. If one exponent is* 1*, the inverse-phase locked gauge makes that row identically* 1 *while preserving a nonzero determinant whenever the coefficient-side determinant is nonzero. If the residual is allowed to depend on both row and column, it reconstructs a consecutive-power matrix whose zeroth row is identically* 1*.*

Formalised: `ResGauObs`, `ResGauObs`, and `ResGauObs`. Thus a residual-blind rank, determinant, or conditioning criterion alone cannot exclude the target. The proposition does not exclude a criterion that also uses an arithmetic coupling identity, and it gives no irrationality criterion for S.

What is open. The reduction converts #249 into the statement that the certificate supply is *unbounded*: certificates exist for arbitrarily large parameters. By Proposition 4.6 this is equivalent to the tail differences being non-integral infinitely often, and equivalently to the Farey gap denominators of Section 4.2 being unbounded: the same obstruction seen as a rational denominator, an eventual binary period, and a flat tail on the lcm cone. That unboundedness is the remaining analytic question, and it is left open. The reduction is real; the closure is not asserted.

5 Formalisation architecture and mathematical lessons

The formal development repeatedly separates a finite, kernel-checkable core from the analytic statement that would have to hold uniformly. For the full-support Erdős–Borwein theorem, the core is a bounded certificate engine assembled from a Bertrand/CRT frame, divisor-pair averaging, and an explicit parameter closure. For #249, the same design principle appears more sharply: eventual binary periodicity predicts an integral tail difference; finite modular arithmetic can refute that prediction at any fixed pair of parameters; and completeness proves that this computation loses no information. Lean therefore makes the open boundary visible: the missing ingredient is not another finite verification but a theorem producing witnesses at unbounded parameters.

The representation of eventual periodicity was itself consequential. Rather than reason about an informal repeating digit string, the development uses the scaled tails R_N and proves that rationality forces their period differences to be integers. Certificate completeness then separates two logically distinct tasks: finite evaluation decides non-integrality at fixed parameters, while an analytic theorem would still be needed to produce such parameters without bound. This separation is what prevents verified computation from being mistaken for the open theorem.

Three changes of coordinates serve different mathematical purposes. The Mersenne–Lambert transform places S next to the Erdős–Borwein constant and exposes the nonnegative primitive-conductor weight. The Möbius-square identity turns the same number into a signed rapidly convergent series. The coprime-pair identity turns it into a probability. Formalising all three and proving that the denominator exclusion transfers between them prevents an informal change of representation from silently changing the claim.

The auxiliary carry modules ask what rationality would force rather than claiming a new irrationality theorem. They give an exact tempered-orbit criterion, Boolean–Möbius coordinates for support series, constraints from residue wraps and reciprocal mass, and a greedy description of the value set. The sublogarithmic coverage theorem adds a further lesson: the elementary fixed-power estimate $\tau(n)^k \leq (k^{2^k})^k n$ becomes useful only after it is transported through the binary tail and composed with the exact carry recurrence. Rationality then forbids long zero windows in f_A, uniformly in the support. In the reciprocal-mass argument, divergent series had to be kept as an explicit alternative rather than passed through the convention that a divergent `tsum` evaluates to zero. Together they rule out several tempting but false proof strategies—for example, rationality need not give a bounded carry alphabet—and identify the kind of infinite Boolean carry path that a future proof would have to exclude. The precise statements and source links are retained in Appendix A; their supporting role is why they are not a second main narrative here.

This architecture is reusable beyond the two named problems: choose a representation in which rationality imposes a rigid tail law, isolate a finite predicate that soundly refutes that law, prove the predicate complete at fixed parameters, and state the remaining uniformity theorem separately. The last step matters as much as the first three. It prevents a large verified finite range from being mistaken for the infinite theorem.

The repository records the corresponding claim-status and change rules in `METHODOLOGY.md`; the mathematical content remains the statements and proofs above.

6 Artefact availability and verification

The Lean 4 proof assistant [20] and Mathlib library [21] check the development. The source, exact informal-to-formal links, and build metadata are maintained in the repository `plectis-lean-erdos249-257`. This paper is pinned to the committed formal-source checkpoint `0e85f7ebadb0f81a26791bc167ce224bd2534f36`; each blue source sigil targets an exact declaration at that revision (for example, the base-2 corollary in Theorem 3.1 resolves to `CerKer`). The package uses `leanprover/lean4:v4.29.1` and the Mathlib revision pinned by `lake-manifest.json`. The exact formal-source revision cited here is `0e85f7ebadb0f81a26791bc167ce224bd2534f36` [23].

There is no `sorry` or `admit` in the package. Finite computations use `decide`, whose proof terms are checked by the Lean kernel; the package does not use `native_decide`. Reproduction requires only the pinned toolchain and the ordinary project build:

```
lake exe cache get
lake build
```

The second command elaborates every declaration and kernel-checks every proof term against the pinned Mathlib dependency. Continuous integration executes the same build. Appendix B gives a compact map of the principal informal statements to declarations. The repository's claim registry (`docs/claims.json`) records each registered claim once, and `scripts/check_release.py` verifies that this paper's source links, the repository documentation, and the release metadata all agree with it, including that every source sigil above resolves to the named declaration at the stated line of the named formal-source checkpoint.

7 Complements and further questions

The formalisation establishes checked results in two directions and leaves two exact frontiers. It verifies the all-integer-base Erdős–Borwein theorem and several structured support cases toward #257. For #249 it proves that a rational value of S would have denominator exceeding Q_0, and it reduces irrationality to an unbounded supply of finite certificates whose fixed-parameter semantics are complete. The unresolved #249 step is precisely that unboundedness; the unresolved #257 step is the passage from the formalised support classes to every infinite support. Neither frontier is softened by the size of the verified artefact.

The remaining questions can be stated without reference to the size of the formal development.

7.1 Exact-arithmetic frontier as of 13 July 2026

Two further reductions put the current proof obligations into scalar exact arithmetic. They do not change the status of either problem.

For #249, let m_t be the canonical adjacent-suffix depth and let d_t be the corresponding residue in $[0, 2^{m_t})$. Define

$$\sigma_t = \min\bigl(d_t - 2^{m_t-5}, (2^{m_t} - 2^{m_t-5}) - d_t\bigr) \in \mathbb{Z}.$$

Proposition 7.1 (canonical strict-jump slack)**.** *The canonical residue lies in the central band* $[2^{m_t-5}, 2^{m_t} - 2^{m_t-5}]$ *if and only if* $\sigma_t \geq 0$*. If nonnegative slack occurs at arbitrarily large strict jumps* $M_t < M_{t+1}$*, then* S *is irrational.*

Formalised: `DiaFreLosBri`, `DiaFreLosBri`, and `DiaFreLosBri`. The proposition is an equivalence and a conditional consumer. It does not prove that σ_t is nonnegative cofinally.

For #257, write r_n for the exact greedy residual of the target $1/2$ and put

$$u_n = 4^n(2r_n - 2^{-n}).$$

The second-channel condition is the rational inequality

$$\frac{1}{6} + \frac{37}{56}2^{-n} \leq \left|u_n - \frac{1}{3}\right|.$$

Proposition 7.2 (rational second-channel reduction)**.** *The second-channel condition is decidable over* $\mathbb{Q}$ *and holds for* $1 \leq n \leq 6$*. If it holds for every* $n \geq 7$*, then* $1/2$ *belongs to the Mersenne achievement set* $\mathcal{A}$*.*

Formalised: `GreAchSet`, `GreAchSet`, and `GreAchSet`. The membership conclusion is itself exact:

$$1/2 \in \mathcal{A} \quad \Longleftrightarrow \quad \text{the canonical greedy orbit omits infinitely many exponents.}$$

This is formalised at `GreAchSet`. The forward implication uses the already formalised irrationality of the full Erdős–Borwein series; the reverse implication uses the absorbing fatal-state criterion. Thus the second-channel hypothesis is one sufficient route to the infinite-skip condition. It does not prove that condition, and membership alone is not the universal #257 theorem.

Exact-arithmetic probes used during proof search found no violation of the second-channel inequality through $n = 4000$, and found nonnegative canonical slack at every tested index through $t = 52$, including all 22 strict lcm jumps in that range. These are finite computations outside the Lean proof authority. In particular, they do not supply either universal quantifier. Their value is diagnostic: the #257 computation points to an invariant of the rational orbit u_n, while the #249 computation points to a sign or recurrence argument for σ_t across fresh prime-power lcm jumps.

Problem 7.3 (Erdős #249)**.** Prove that

$$\sum_{n\geq 1} \frac{\varphi(n)}{2^n}$$

is irrational, without imposing a bound on the denominator of a possible rational representation.

Problem 7.4 (unbounded certificate supply)**.** Produce certified non-integrality witnesses at unbounded parameters, equivalently along the lcm diagonal. This is the open hypothesis in Corollary 4.4; the verified scales in Example 4.8 form a finite initial set and do not establish it.

Problem 7.5 (universal Erdős #257)**.** Prove that

$$\sum_{n\in A} \frac{1}{2^n - 1}$$

is irrational for every infinite set $A \subseteq \mathbb{N}$, rather than only for the support families in Theorem 3.2.

On the #257 side, rationality would impose two complementary forms of rigidity: divisor-coverage zero windows have length $o(\log N)$, whereas an infinite support forces the associated positive carry states to be unbounded. These statements constrain different objects and do not yet exclude a rational infinite-support value, but together they narrow the structure that any such value would have to exhibit.

The main lesson from formalisation is the alignment of representations and proof obligations. The Lambert, Möbius, probabilistic, periodic-tail, and carry descriptions are not competing stories: each makes a different rationality obstruction explicit. A future advance must exploit one of those obstructions uniformly, not merely extend the finite range already checked. The public package therefore records exact implications, finite results, and negative results without treating corpus volume as mathematical evidence or claiming that an unchecked composition closes either open problem.

Statements and declarations

Artefact and data availability. The pinned formal-source revision `0e85f7ebadb0f81a26791bc167ce224bd2534f36` contains the Lean sources, pinned toolchain, Mathlib manifest, and generated certificate files used by the source sigils. The claim registry and this manuscript are repository navigation surfaces rather than proof authority; Section 6 gives the verification route.

Use of AI assistance. Large-language-model assistants were used during development for proof drafting, refactoring, and prose editing. Every mathematical claim rests on the published Lean source, in which each proof term is checked by the Lean 4 kernel against the pinned Mathlib, with no `sorry`, `admit`, additional axioms, or `native_decide`.

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A Auxiliary rationality criteria from binary carry systems

Five further modules provide additional criteria. They prove no new irrationality. Instead they pin down, in machine-checked form, what a rational value of the series in question would have to look like: as an integer carry orbit, as a Möbius-inverted support coordinate, as a residue-wrap cycle modulo the odd part of the denominator, and as a point of a greedy subset-sum geometry. Everything in this section is an equivalence or an unconditional constraint on the rational side; nothing excludes a rational value of either open series, and each module states that boundary in its header. The prior-art boundary is theorem-family-specific. The integral carry recurrence has a direct antecedent in Wang–Grau Ribas [16]; the strict-tail geometry is recorded by Kovač–Tao [15]; Möbius inversion, repetend algebra, and divisor averaging are classical (see, for example, Apostol [7]). We make no claim of mathematical priority for the converse/rigidity, certificate-normal-form, or coupled reciprocal-mass/collision statements collected here.

A.1 Tail-orbit rigidity: a rationality criterion

For a coefficient sequence $c : \mathbb{N} \to \mathbb{N}$ with $c(n) \le n$, write

$$X_c = \sum_{n\ge1} \frac{c(n)}{2^n}, \qquad T_c(N) = \sum_{j\ge1} \frac{c(N+j)}{2^j},$$

the series and its scaled tail after N binary digits. Call an integer sequence u a *tempered orbit* for c with multiplier $v \ge 1$ if

$$u(N+1) = 2\,u(N) - v\,c(N+1) \ \text{ for all } N, \qquad \text{and} \qquad u(N)/2^N \to 0.$$

The recurrence is exact binary long division against the coefficient stream; the boundary condition forbids the homogeneous solution. Wang and Grau Ribas use the integral carry recurrence forced by rationality for the weighted binary special case $c(n) = nd_n$ [16]. The module abstracts that forward mechanism to arbitrary nonnegative $c(n) \le n$; its explicit converse and subexponential-orbit rigidity are the local additions, with no priority claim attached. Wang–Grau Ribas's positive-density theorem and its Erdős #260 corollary are not used or formalised here.

Theorem A.1 (tail-orbit rigidity)**.** *Let $c(n) \le n$ for all n. Every tempered orbit is the scaled tail, $u(N) = v\,T_c(N)$ for all N; and X_c is rational if and only if a tempered orbit exists for some multiplier $v \ge 1$.*

Formalised: `GenTaiOrbRig` (rigidity) and `GenTaiOrbRig` (the criterion), restated in Mathlib's irrationality vocabulary at `GenTaiOrbRig`. The engine is one observation: a real orbit with $d(N+1) = 2\,d(N)$ and $d(N) = o(2^N)$ vanishes identically (`GenTaiOrbRig`).

The tempered boundary is essential: the homogeneous parasite $N \mapsto 2^N$ can be added to any orbit without disturbing the recurrence, so positivity of an orbit certifies nothing by itself, and positivity is nowhere advertised as an equivalent criterion. The linear-growth hypothesis covers both constants introduced in Section 1: $\varphi(n) \le n$ for the #249 coefficients, and $f_A(n) \le \tau(n) \le n$ for the #257 support coefficients $f_A(n) = \#\{a \in A : a \mid n\}$. The modules below instantiate the criterion on supports; the totient instantiation is not taken here.

Proposition A.2 (finite-level carry anti-compression)**.** *Suppose a tempered integral carry orbit for the totient sequence has positive multiplier. If the canonical level-e totient kernel family is linearly independent, then the span of its canonical carry sections has dimension at least $2^e - 1$. Consequently, an all-level independence producer together with rationality of the totient coefficient series would force one tempered carry orbit with unbounded finite-level section ranks.*

Formalised: `TotCarKerRig` and `TotCarKerRig`. The all-level independence producer is not supplied here. Thus the proposition records an exact conditional anti-compression consequence, not an irrationality criterion for S.

A.2 Boolean–Möbius carry coordinates (the #257 direction)

At base 2 the support series of Section 3 is itself a binary coefficient series, $\sum_{n\in A} 1/(2^n-1) = X_{f_A}$ (BooMobCar). Two exact coordinate changes then turn a hypothetical rational value into a support-free object. The divisor transform and Möbius inversion below are classical Lambert-series infrastructure [7, 8, 9]; the contribution here is their Lean-checked composition with the tempered carry interface.

The signed identity $L(\mu) = 1/2$ does not by itself provide a Boolean support for the target value. If the negative Möbius indices are selected and the positive indices (apart from 1) are discarded, the selected support sum is exactly $1/2$ plus the positive Möbius tail, hence exceeds $1/2$ by at least $1/63$ from the index 6. This closes one sign-truncation route only; it does not exclude another infinite Boolean support with value $1/2$. Formalised: MobSigSupNoGo and MobSigSupNoGo.

Proposition A.3 (Boolean Möbius inversion, both directions)**.** *On positive integers, $f_A = \mathbf{1}_A * \zeta$ and $\mu * f_A = \mathbf{1}_A$, where $\mathbf{1}_A$ is the indicator of A. Conversely, every integer-valued arithmetic function f whose Möbius transform is Boolean, $(\mu * f)(n) \in \{0,1\}$ for all $n \geq 1$, satisfies $f = f_B$ for the support $B = \{n \geq 1 : (\mu * f)(n) = 1\}$ selected by that transform.*

Formalised: BooMobCar, BooMobCar, and, with no search or finiteness hypothesis, the converse BooMobCar; the selected support is BooMobCar.

On the carry side, Theorem A.1 applies at $c = f_A$: the support series is rational exactly when a tempered carry orbit for f_A exists (BooMobCar); a displayed fraction p/q corresponds to an orbit U with multiplier q and initial value $U(0) = p$ (BooMobCar); and exact division recovers the coefficient from the orbit, $\big(2U(N) - U(N+1)\big)/q = f_A(N+1)$ (BooMobCar). The elementary divisor-pair bound $\tau(n) \leq 2\lfloor\sqrt{n}\rfloor$ (BooMobCar) confines the tails to a square-root strip, $T_{f_A}(N) \leq 2\sqrt{N} + 4$ (BooMobCar), hence every orbit state to $0 < U(N) \leq q\,(2\sqrt{N}+4)$ (BooMobCar, BooMobCar); and the strip is strong enough to re-derive the tempered boundary, so the analytic side condition can be traded for one inequality. The two coordinate changes compose.

Theorem A.4 (certificate-level equivalence)**.** *Fix an integer p and an integer $q \geq 1$. There is a nonempty support A of positive integers with $\sum_{n\in A} 1/(2^n-1) = p/q$ if and only if there is an integer sequence U with $U(0) = p$ such that: every $U(N)$ is positive; $U(N) \leq q\,(2\sqrt{N}+4)$; q divides $2U(N) - U(N+1)$ for every N; and the Möbius transform of the quotient sequence $\big(2U(N) - U(N+1)\big)/q$ is Boolean. The support is not guessed: it is reconstructed from the certificate by Proposition A.3.*

Formalised: BooMobCar, with the certificate packaged as BooMobCar and the quotient normalized at index zero (BooMobCar).

Example A.5 (the support $\{2,3\}$)**.** The support $\{2,3\}$ has value $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$ (BooMobCar), its orbit is the pure period-six cycle $10, 20, 19, 17, 13, 26$ with multiplier 21 (BooMobCar), and the Möbius transform of the carry quotient recovers exactly $\{2,3\}$ (BooMobCar).

These are coordinates, not by themselves a strategy: nothing here excludes infinite Boolean carry paths, and no finite search is promoted to a completeness argument.

A.3 Sublogarithmic divisor coverage

The coefficient $f_A(n)$ counts the elements of A dividing n. Say that a zero window of length h starts after $c + N$ when $f_A(c + N + j + 1) = 0$ for every $0 \leq j < h$. Thus no integer in that interval is divisible by an element of the support.

Theorem A.6 (sublogarithmic zero windows)**.** *Let A contain a positive integer and suppose*

$$\sum_{a\in A}\frac{1}{2^a-1}=\frac{p}{2^c v},\qquad p\in\mathbb{Z},\quad c\in\mathbb{N},\quad v\geq 1.$$

For every $\varepsilon>0$ there is a constant $B\geq 0$, depending only on ε, c, and v, such that every zero window after $c+N$ satisfies

$$h\leq \varepsilon\log_2(N+1)+B,\qquad \log_2 x:=\frac{\log x}{\log 2}.$$

The bound is uniform in A, p, N, and h.

Formalised: `SubDivCov`.

The proof first establishes, for each integer $k\geq 2$, the explicit divisor estimate

$$\tau(n)^k\leq \left(k^{2^k}\right)^k n.$$

It transports the resulting $n^{1/k}$ bound through the binary tail and then uses the exact carry recurrence across a zero window. Raising the resulting inequality to the kth power absorbs the occurrence of h on the right and leaves a coefficient $1/(k-1)$ in front of the binary logarithm. Choosing k after ε gives the theorem. This controls gaps in divisor coverage; it does not assert a subpower bound for the carry state itself.

A.4 Residue wraps, reciprocal mass, and unbounded carries

Suppose now the support series equals $p/(2^c v)$ with v odd and A containing a positive element. Rationality makes the scaled tails integers: $u(N)=v\,T_{f_A}(c+N)$ is a positive integer, satisfies the carry recurrence $u(N+1)+v\,f_A(c+N+1)=2\,u(N)$, and reduces modulo v to the doubling residues $r_N=p\,2^N \bmod v$ (`RatSupCarSke`). Doubling a least residue either stays under v or wraps past it exactly once, so each step emits a wrap digit $k_N=\lfloor 2r_N/v\rfloor\in\{0,1\}$, and over any cycle length h with $2^h\equiv 1 \pmod v$ the repetend identity holds:

$$\sum_{N<h} r_N \;=\; v\cdot w,\qquad w=\sum_{N<h} k_N,$$

with $w\geq 1$ when p is coprime to $v>1$. Formalised: `RatSupCarSke` and `RatSupCarSke`; residues, digits, and counts are `RatSupCarSke`, `RatSupCarSke`, `RatSupCarSke`.

Proposition A.7 (one-wrap classification)**.** *Let $v>1$ be odd, p coprime to v, and $h\geq 1$ with $2^h\equiv 1 \pmod v$. If the cycle of p wraps exactly once, then $v=2^h-1$ and the starting residue is a power of two: $r_0=2^a$ for some $a<h$.*

The one-wrap cycles are thus exactly the Mersenne repetends $2^a/(2^h-1)$. Formalised: `RatSupCarSke`, from the generic closed-cycle form `RatSupCarSke`, and instantiated at the true order $h=\operatorname{ord}_v(2)$ (`RatSupCarSke`, with `RatSupCarSke`). The classification is algebraic; a finite check of 446 coprime starting residues across twelve modulus/order rows is retained as kernel-checked validation, not as the proof (`RatSupCarSke`, `RatSupCarSke`, `RatSupCarSke`).

The analytic bridge is Cesàro averaging. Write $\rho(A)=\sum_{a\in A}1/a$ for the reciprocal mass (`RatSupCarSke`). When the reciprocal family is summable, the mean of $f_A(1),\ldots,f_A(N)$ converges to $\rho(A)$ — the density of the multiples of a, summed over the support — and the mean of the scaled tails has the same limit (`RatSupCarSke`, `RatSupCarSke`). The divergent case is carried as an explicit disjunction, not through the convention that a divergent `tsum` is zero (`RatSupCarSke`). Each tail state splits exactly as $u(N)=v\,e_N+r_N$ with integral excess $e_N=\lfloor u(N)/v\rfloor\geq 0$ (`RatSupCarSke`).

Proposition A.8 (order-wrap lower bound, with exact excess mean)**.** *Let A and $p/(2^c v)$ be as above with $v > 1$ odd, and let w be the wrap count of p over one cycle of length $h = \mathrm{ord}_v(2)$. Then $\sum_{a\in A} 1/a$ diverges or $\rho(A) \geq w/h$; if moreover p is coprime to v, then $w \geq 1$, so $\rho(A) \geq 1/\operatorname{ord}_v(2)$. In the summable case the bound is the periodic part of an exact identity: the Cesàro mean of the excess e_N converges automatically, and*

$$\rho(A) \;=\; \frac{w}{h} \;+\; \lim_{N\to\infty} \frac{1}{N} \sum_{n<N} e_n.$$

Formalised: RatSupCarSke, RatSupCarSke, the form retaining the divergent alternative RatSupCarSke, and the excess mean RatSupCarSke (fraction-facing shifted form RatSupCarSke, named-limit form RatSupCarSke). A rational value thus ties the odd part of its denominator to the sparsity of its support: a summable support with small $\rho(A)$ can only display denominators whose odd part has small multiplicative order of 2.

Corollary A.9 (collision strengthening at common multiples)**.** *In the summable case, let $F \subseteq A$ be finite and let L be a common multiple of F. Then*

$$\rho(A) \;\geq\; \frac{w}{h} \;+\; \frac{\lceil (|F|-1)/2 \rceil}{L}.$$

In particular, an infinite support whose value is a dyadic rational $p/2^c$ has $\sum_{a\in A} 1/a$ divergent or $\rho(A) > 1$.

The mechanism is a collision: at every positive multiple of L at least $|F|$ support divisors land on the same index, so $f_A \geq |F|$ there, while the exact carry equation $f_A(n) + e_n = k_{n-1} + 2e_{n-1}$ with $k_{n-1} \leq 1$ lets one step absorb at most one unit — so the preceding excess must spike to at least $\lceil (|F|-1)/2 \rceil$, and the spikes recur with density $1/L$. Formalised: RatSupCarSke (denominator-shifted form RatSupCarSke), the carry equation RatSupCarSke, and the dyadic corollary RatSupCarSke (via RatSupCarSke).

Theorem A.10 (carry states are unbounded over infinite supports)**.** *Let A be infinite with value $p/(2^c v)$, $v \geq 1$. Then the positive integer carry state $u(N) = v\,T_{f_A}(c+N)$ is unbounded: for every B there is an N with $u(N) > B$.*

The proof picks $2B+1$ positive support elements; at a common multiple L of all of them, the local inequality $1 + v\,|F| \leq 2\,u(L-c-1)$ pushes the state past B. Formalised: RatSupCarSke, from RatSupCarSke and the local inequality RatSupCarSke; the $\{2,3\}$ fixture revalidates it at sixty-six common multiples (RatSupCarSke). So a rational value over an infinite support cannot run on a bounded carry alphabet. That closes finite-state readings of the carry system; it does not close the problem. Nothing in the constraints above prevents an infinite support from satisfying every one of these constraints. The repetend identities and divisor averages used here are classical. We make no claim of mathematical novelty for the coupled reciprocal-mass bounds, collision strengthening, or global unboundedness statement.

A.5 Greedy geometry of the #257 value set

The fourth module studies the set of all values a base-2 support series can take. For $n \geq 1$ put $w_n = 1/(2^n - 1)$, let $T(n)$ be the mass strictly after exponent n, and let

$$\mathcal{A} \;=\; \Big\{ \sum_{n\in A} w_n \;:\; A \subseteq \mathbb{N},\; 0 \notin A \Big\},$$

the *Mersenne achievement set*, the exponent 0 being normalised away as analytically invisible. Coding by supports of positive exponents is literally the formal support series (GreAchSet; the set is GreAchSet). In this language the base-2 #257 statement reads: the rational members of $\mathcal{A}$ are exactly the finite subset sums. Nothing below decides that. Kovač and Tao verify, for every fixed integer $t \geq 2$, that

$$\sum_{\ell > n} \frac{1}{t^\ell - 1} < \frac{1}{t^n - 1},$$

and deduce distinct Lambert subsums and a Cantor set [15]. At $t = 2$ this is the relevant Lambert-series instance of Kakeya's classical strict-tail criterion for subsum sets [10]. The local contribution is the executable Lean interface, quantitative enclosure, and finite non-membership certificates over that known geometry; it does not attribute those refinements to the cited sources or settle #257.

Theorem A.11 (superincreasing geometry and greedy membership)**.** *For every $n \geq 1$ the weight dominates the whole tail after it, $T(n) < w_n$, with the two-scale gap $w_n - T(n) = \frac{2}{3}4^{-n} + O(8^{-n})$ and explicit valid O-constant 3. Consequently a real x belongs to $\mathcal{A}$ exactly when $x \geq 0$ and every greedy residual of x is at most the remaining tail; and normalised support coding is injective, so each member of $\mathcal{A}$ has a unique support, which the greedy recursion recovers bit by bit.*

Formalised: GreAchSet, GreAchSet (explicit bound GreAchSet), GreAchSet, and GreAchSet, transferred to the kernel series at GreAchSet. The nonnegativity guard in the membership criterion is necessary: without it a negative target would survive every level while lying outside $\mathcal{A}$.

The binary coding also permits the global geometry to be checked directly.

Proposition A.12 (fat-Cantor geometry)**.** *The achievement set $\mathcal{A}$ is compact, perfect, totally disconnected, and nowhere dense. Its Lebesgue measure is exactly one.*

Formalised: GreAchSet, GreAchSet, GreAchSet, and GreAchSet. Kovač–Tao supply the strict-tail Cantor-set context cited above. The formalisation records these exact properties without making a novelty or priority claim for them.

Proposition A.13 (exact rational runs and finite non-membership certificates)**.** *The greedy recursion executed in exact rational arithmetic agrees step by step with the real recursion under casting. A finite decidable certificate — the exact rational residual at some level is at least an exact rational upper enclosure of the remaining tail — soundly proves non-membership in $\mathcal{A}$; the level-one instance shows $3/4 \notin \mathcal{A}$. And a rational number already known to lie in $\mathcal{A}$ has computable support bits.*

Formalised: GreAchSet; GreAchSet, with the certificate GreAchSet and the enclosure GreAchSet; GreAchSet; and GreAchSet. The certificates are one-sided: survival through any finite depth proves nothing about membership, and no decidability of membership in $\mathcal{A}$ is asserted. Neither the global geometry nor the finite death certificates exclude rational values for either open series.

B Index of principal declarations

Each row links to the declaration at source revision 0e85f7ebadb0f81a26791bc167ce224bd2534f36.

informal statement	linked Lean source
$\sum 1/(b^n - 1)$ irrational, all $b \geq 2$	CerKer (CertificateKernel)
Erdős–Borwein E irrational	CerKer (CertificateKernel)
factorial support	CerKer (CertificateKernel)
power-of-two support	CerKer (CertificateKernel)
pairwise-coprime supports	CerKer (CertificateKernel)
$L(\mu) = 1/2$	MerLamLad (MersenneLambertLadder)
$L(\varphi) = 2$	MerLamLad (MersenneLambertLadder)
positive lift $L(A) = S$	CerKer (CertificateKernel)
Möbius-square lens	CerKer (CertificateKernel)
$L_2(\mathbf{1})$ and $L_2(\varphi)$ gcd moments	GcdMomCal, GcdMomCal (GcdMomentCalculus)
Stern–Brocot cylinder recursion and rate	GcdMomCal, GcdMomCal (GcdMomentCalculus)
$S = \frac{1}{2} + \Pr(\gcd = 1)$	CerKer (CertificateKernel)
$S \neq p/q$ for $q \leq Q_0$	CerKer (CertificateKernel)
tail-period reduction	TotTaiPerKil (TotientTailPeriodKiller)
diagonal reduction	LcmDiaRed (LcmDiagonalReduction)
cone flatness from rationality	LcmConFla (LcmConeFlatness)
certificate completeness	LcmConFla (LcmConeFlatness)
joint cone refuter	LcmConNon (LcmConeNonflat)
verified finite instances	TotTaiPerKil (TotientTailPeriodKiller)
full-target diagonal reduction	DiaPinDec (DiagonalPincerDecomposition)
squared-Mersenne enclosure	SquMerDiaEnc (SquaredMersenneDiagonalEnclosure)
exact Mersenne-shadow denominator	MerShaDenGro (MersenneShadowDenominatorGrowth)
fresh-loss projection reduction	DiaFreLosBri (DiagonalFreshLossBridge)
prime-adjunction diamond no-go	FulTarPriAdjNoGo (FullTargetPrimeAdjunctionNoGo)
scalar-localisation height obstruction	AdeHeiObs (AdelicHeightObstruction)
square-CRT correction suppression	SquCRTCub (SquareCRTCube)
signed dyadic-moment algebra	SigQMomObs (SignedQMomentObstruction)
dyadic totient certificate interface	TotMahDef (TotientMahlerDefect)
residual-gauge obstruction for monomial minors	ResGauObs (ResidualGaugeObstruction)
tail-orbit rationality criterion	GenTaiOrbRig (GenericTailOrbitRigidity)
Boolean carry certificate $\Leftrightarrow$ support fraction	BooMobCar (BooleanMobiusCarry)
one-wrap classification	RatSupCarSke (RationalSupportCarrySkeleton)
order-wrap mass bound	RatSupCarSke (RationalSupportCarrySkeleton)
unbounded carry states	RatSupCarSke (RationalSupportCarrySkeleton)
sublogarithmic zero windows	SubDivCov (SublogDivisorCoverage)
greedy membership criterion	GreAchSet (GreedyAchievementSet)
Mersenne achievement set: measure one and nowhere dense	GreAchSet, GreAchSet (GreedyAchievementSet)

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